Mudit Mehrotra

Senior Software Engineer | Full-Stack & AI/ML Engineer

muditmehrotra9999@gmail.com ✉
+91-7906537025 ☎
linkedin.com/in/mudit-mehrotra999 in
github.com/mudit9999 </>

Full-Stack AI/ML Engineer with 5+ years of experience specializing in building end-to-end web applications, intelligent conversational agents, and production-grade, scalable RAG architectures. Experienced in bridging modern frontend and backend engineering (MERN, Python, FastAPI) with LLMs (LangChain, LangGraph) to deliver high-concurrency, real-time AI solutions.

WORK EXPERIENCE

AI/ML Engineer — Tiger Analytics
05/2023 – Present

Retail Sales & Operations Intelligence Platform (Apparel Retail — Gap Inc.)

- Architected and scaled an enterprise retail analytics platform serving **~4,000 daily active users** (Store Managers, District Managers, and Executives) to track real-time sales and operational KPIs.
- Engineered **Insights Assist**, a single-agent **LangChain** solution that dynamically extracts location- and time-bound operational metrics to generate automated executive summaries and actionable recommendations.
- Developed a production-grade, multi-agent, multi-tool conversational chatbot using **LangGraph** to resolve store- and district-level analytical queries with end-to-end response latency under **~10–20 seconds**.
- Implemented strict **RBAC (Role-Based Access Control)**, **Model Armor**, and LLM safeguard guardrails, alongside multi-modal **Text-to-Speech (TTS)**, **Speech-to-Text (STT)**, and contextual follow-up question generation.

Procurement & Supply Chain Management Tool (PepsiCo — US & Canada Plants)

- Built and automated a comprehensive **Procurement & Supply Tool** managing raw material (Potato) procurement and supply across US and Canada manufacturing plants, covering end-to-end demand **planning, forecasting, and actual supplied material tracking**.
- Developed **50+ high-performance backend REST APIs using FastAPI** and optimized **30+ complex SQL Server** business logic procedures, managing seamless migrations across Dev, QA, and Production environments.

Shared Services & Enterprise Budgeting Applications (Pharma & Enterprise)

- Built a full-stack **Shared Services Request Application using the MERN Stack** (PostgreSQL, Nest.js, React.js) for a leading Pharma client, boosting Operational Excellence team efficiency by **50%** and enabling data-driven budget allocation by location and request type.
- Developed a centralized **Headcount & Budget Management Tool** to streamline budget planning and multi-stage approval workflows for new joiners and internal movers.
- Collaborated with project architects on system design, peer task allocation, internal AI/engineering initiatives, and UAT defect resolution to ensure zero-defect production deliveries.

System Engineer — Tata Consultancy Services (TCS)
02/2021 – 04/2023

- Engineered reusable, modular workflow automation packages that standardized deployment across multiple client environments, significantly reducing engineering effort and operational maintenance costs.
- Developed custom **Risk Management web applications** backed by structured data lists, eliminating client dependency on manual spreadsheets and improving data governance.

TECHNICAL SKILLS

AI / ML & GENAI

LangChain | LangGraph | Multi-Agent Systems | Scalable RAG | LLM Guardrails & Model Armor | RBAC

FULL-STACK & BACKEND

Python | FastAPI | React.js | Node.js | Nest.js | REST APIs | JavaScript

DATABASES, CLOUD & TOOLS

SQL Server | Postgres | Azure Data Factory | Git / GitHub | Power BI

EDUCATION

B.Tech in Information Technology
Ajay Kumar Garg Engineering College (A.K.T.U.)
07/2016 – 09/2020
• Member of Google Developers Club (GDC)

ACHIEVEMENTS

Smart India Hackathon (SIH) 2018
National Finalist of Smart India Hackathon 2018

TCS CodeVita Season 8
Secured Global Rank 4520

CERTIFICATIONS

Microsoft Certified: Data Analyst Associate
Power BI Data Analytics & Visualization (PL-300)

LANGUAGES

English & Hindi
Full Professional Proficiency

INTERESTS

Problem Solving | Cricket | Music